

Stator Teeth Shape Design for Torque Ripple Reduction in Surface-Mounted Permanent Magnet Synchronous Motor

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Abstract — This paper presents the method on the stator teeth shape design in Surface-Mounted Permanent Magnet Synchronous Motor (SPMSM) for reducing torque ripple. In order to examine the effect of the stator teeth shape to torque ripple, three models of the SPM motor which have the different stator teeth shapes are selected, and then FEM analysis is carried out to compare their characteristics. Then, a method about specifying a design parameter to be able to effectively reduce torque ripple is developed. By use of this method, the optimized model can be achieved rapidly and reliably. The torque ripple is reduced by cogging torque which can decrease torque value of maximum points and increase that of minimum points.

I. INTRODUCTION

Permanent magnet (PM) motors have applied widely in more and more applications due to definite advantages, such as high efficiency, high power density, high torque-to-current ratio and high torque-to-volume ratio [1], [2]. However, the torque ripple caused by harmonic components of radial flux density has undesirable effects. For instance, especially in high-precision, low-speed position control system, such as electric power steering (EPS) systems, or direct driven electric systems, where there is no reduction gear, torque ripple reduction is the important issue [3]-[5].

For the case of SPM type motor, in contrast to IPM type motor, the combination of cogging torque and alignment torque yield the total torque, excluding reluctance torque, which is generated by the rotor saliency. The ripple of the alignment torque is generated by harmonic components of current and back electromotive force (EMF). In contrast, independently of current in stator coil, the cogging torque affected by interacting between the permanent magnets and the stator shape is pulsating torque induced by the force to keep reluctance at the minimum. In order to minimize cogging torque and torque ripple, various techniques have been researched [3]-[16]. Torque ripple reduction problem in PM motors can be handled through motor design and control point of view [14]-[16]. Generally, torque ripple can be reduced by minimizing cogging torque [5], [6]. However, cogging torque which can compensate the ripple of alignment torque is helpful to reduce torque ripple.

In this paper, three models of SPM motor which have different stator teeth shapes are compared, and then a method to design stator teeth shape for reducing torque ripple is developed.

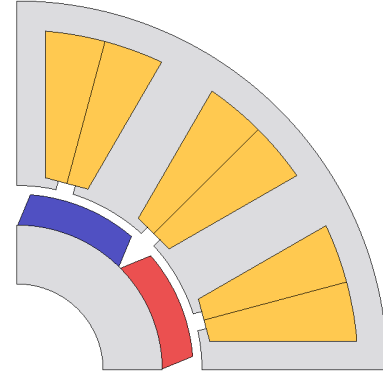


Fig. 1. 8-pole/12-slot SPM motor

TABLE I
DESIGN SPECIFICATIONS OF SPM MOTOR

	Design Parameters	Specification	Unit
General	No. of Poles and slots	8 / 12	
	No. of Phases	3	
	Air-gap	1	mm
	Stack length	30	mm
Stator	Outer Diameter	42	mm
Rotor	Outer Diameter	20	mm
	PM Property	N38UH	

II. CHARACTERISTICS ANALYSIS WITH STATOR TEETH SHAPE IN SPMSM

A. Design Specification and Configuration

Fig. 1 shows the cross-section of 8-pole, 12-slot SPM motor, where the air gap is 1 mm and outer diameters of stator and rotor are 42 mm and 20 mm, respectively, and Table I describes the design specification of SPM motor, studied in this paper.

B. Characteristics Analysis with Stator Teeth Shapes

Fig. 2 shows the three cross-section of the SPM motors which have different shapes of stator teeth. The first model,

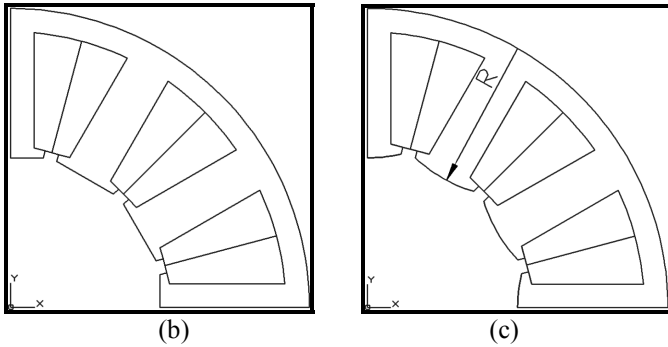
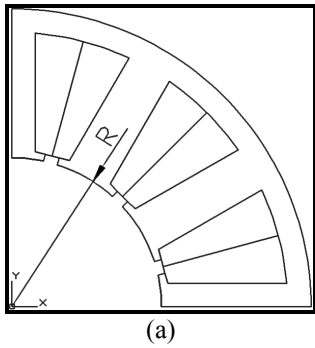


Fig. 2. Cross-section shapes of analysis models. (a) Model 1. (b) Model 2. (c) Model 3.

TABLE II
CHARACTERISTICS OF SPM MOTORS BY STATOR TEETH SHAPES

	Model 1	Model 2	Model 3	Unit
Average Torque	3.85	3.66	3.47	Nm
Cogging Torque	0.244	0.130	0.051	Nm
Back-EMF THD	2.65	1.37	0.86	%
Torque ripple	8.64	4.49	3.65	%

shown in Fig. 2(a), is the initial model. In the second model, the top of the stator teeth is flat. In the third model, the air-gap becomes large toward the stator teeth tip. In order to observe the effect of the stator teeth shape in the characteristics of the SPM motors, the models shown in Fig. 2 should have the same rotor configuration and minimum air-gap length, as shown in Fig. 1.

Table II represents the comparison results of the characteristics of average torque, cogging torque, back-EMF total harmonic distortion (THD), and torque ripple analyzed by FEM according to the stator teeth shapes. Note that these results are achieved by applying the same current density and coil space factor for each model. As a result of comparison, cogging torque, torque ripple and back-EMF THD can be reduced by enlarging the air-gap at the stator teeth tip, where the values of each of those in model 3 are 79.1%, 57.8%, and 67.5% lower than in those of model 1, respectively. However,

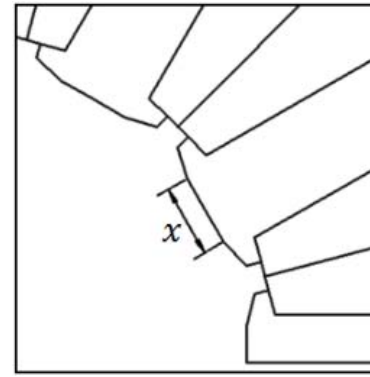


Fig. 3. Design parameter for shaping stator teeth of analysis model

a decline in average torque, which is undesirable, is also occurred. The average torque of model 3 is 9.9% lower than that of model 1.

III. DESIGN FOR TORQUE RIPPLE REDUCTION IN SPMSM

A. Design Parameter and Constraints

The stator teeth shape is described with a design parameter x in Fig. 3. The setting of the design parameter x is based on the results of the characteristics analysis of model 2 and model 3. These models have lower cogging torque and torque ripple than model 1.

Even if the value of the design parameter x is changed, the constant slot open size, slot area, and minimum air-gap length between stator and rotor can be maintained. Therefore, the characteristics comparison of the SPM motors to observe the effect of the different shapes of the stator teeth can be carried out. In addition, because there is one design parameter to optimize the stator teeth shape, the result can be obtained rapidly and reliably. The design parameter and constraints are given as follows:

• Constraints :

$$\begin{aligned} \text{Cogging torque} &\leq 0.1 \text{ Nm} \\ \text{Torque ripple} &\leq 3 \% \\ \text{Average torque} &\geq 3.45 \text{ Nm} \\ \text{THD} &\leq 1 \% \end{aligned}$$

• Design parameter :

$$0 \text{ mm} \leq x \leq 8 \text{ mm}$$

B. Design Results

Fig. 4 shows the cogging torque, torque ripple, average torque and back-EMF THD analysis results according to the value of the design parameter x . The results show that the average torque and back-EMF THD is increased when the value of the design parameter x increases, and as the design parameter x increases from 1.4 mm to 6.4 mm, the cogging torque is increased but the torque ripple is decreased.

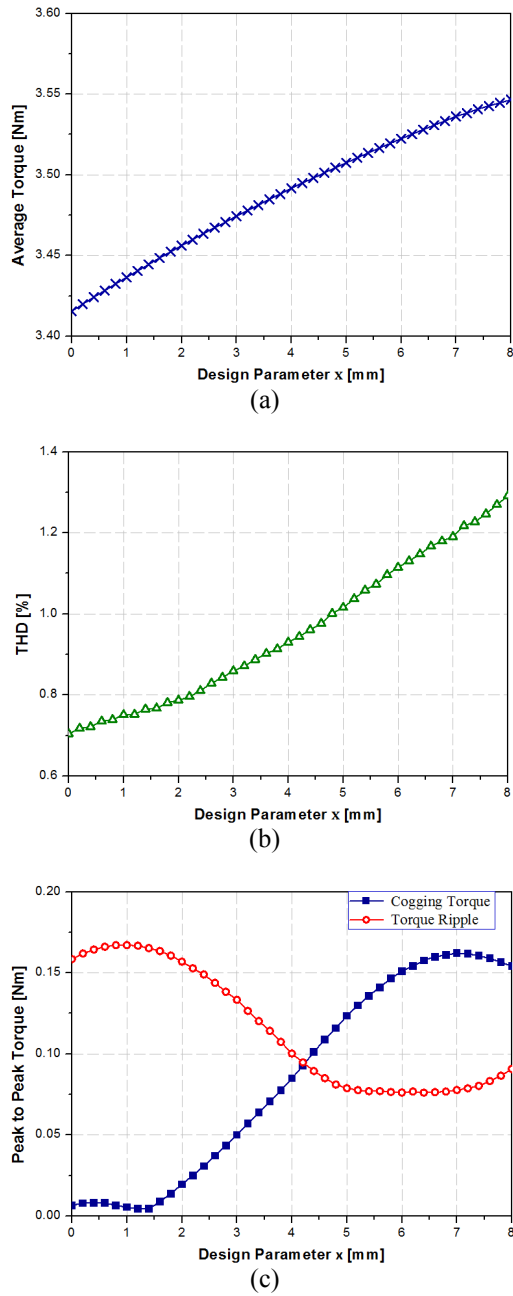


Fig. 4. Analysis results according to the design parameter. (a) Average torque. (b) Back-EMF THD. (c) Cogging torque and torque ripple.

Fig. 5 shows the waveforms of cogging torque, alignment torque and total torque according to the electrical angle when the design parameter x is 1.4 mm, 3.0 mm, 4.6 mm and 6.4 mm, and the quantitative numerical values of the SPM motor characteristics changed by design parameter x are presented in Table III. Total torque is achieved by adding cogging torque to alignment torque. As the value of the design parameter x increases from 1.4 mm to 6.4 mm, so do the ripple of the alignment torque and the cogging torque. However, the ripple of the alignment torque is compensated by adapting cogging torque. Therefore, the cogging torque increases, the torque ripple decreases in the limited region.

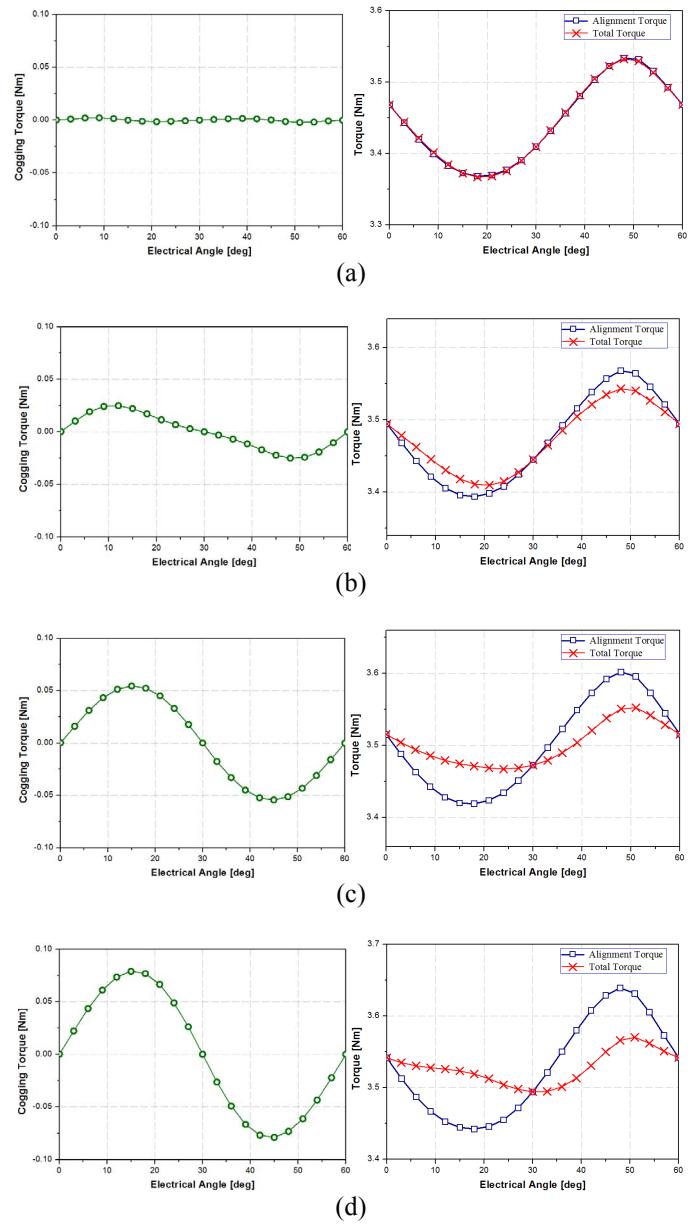


Fig. 5. Analysis results of cogging torque, alignment torque and total torque. (a) $x = 1.4$ mm. (b) $x = 3.0$ mm. (c) $x = 4.6$ mm. (d) $x = 6.4$ mm.

TABLE III
CHARACTERISTICS OF SPM MOTOR BY DESIGN PARAMETER x

	$x = 1.4$ mm	$x = 3.0$ mm	$x = 4.6$ mm	$x = 6.4$ mm	Unit
Average Torque	3.44	3.47	3.50	3.53	Nm
Cogging Torque	0.004	0.050	0.109	0.158	Nm
Back-EMF THD	0.77	0.86	0.98	1.15	%
Alignment torque ripple	0.165	0.174	0.183	0.197	Nm
Torque ripple	4.80	3.84	2.43	2.16	%

As the analysis results, in order to satisfy the condition of constraints, design parameter x should be between 4.0 mm and 4.2 mm. As the minimum value of the cogging torque and the torque ripple cannot be achieved from the same value of the design parameter x , the design parameter x should be carefully determined to design motor reliably.

IV. CONCLUSION

This paper presents a method on how to design the stator teeth shape in SPM motor to reduce torque ripple. The characteristics results of the three models which have different stator teeth shapes are compared, and then the design parameter to be able to effectively reduce torque ripple is applied to design stator teeth shape design. Applying one design parameter in motor shape design is not only simple and quick but also reliable. The comparisons of analysis results show that when the design parameter increases, the back-EMF THD is increased, and therefore the ripple of the alignment torque is also increased. However, as the ripple of the alignment torque is compensated by adapting cogging torque, total torque ripple can be reduced even if cogging torque is increased. The minimum points of the cogging torque and torque ripple is not at the same point, the value of the design parameter which can satisfy the constraints should be selected carefully.

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REFERENCES

- [1] T. Fan, Q. Li, and X. Wen, "Development of a high power density motor made of amorphous alloy cores," *IEEE Transactions on Industrial Electronics*, Vol. 61, No. 9, pp. 4510-4518, September 2014.
- [2] M. J. Kim, S. Y. Cho, K. D. Lee, J. J. Lee, J. H. Han, T. C. Jeong, W. H. Kim, D. H. Koo, and J. Lee, "Torque density elevation in concentrated winding interior PM synchronous motor with minimized magnet volume," *IEEE Transactions on Magnetics*, Vol. 49, No. 7, pp. 3334-3337, July 2013.
- [3] W. Fei, and P. C. K. Luk, "Torque ripple reduction of a direct-drive permanent-magnet synchronous machine by material-efficient axial pole pairing," *IEEE Transactions on Industrial Electronics*, Vol. 59, No. 6, pp. 2601-2611, June 2012.
- [4] W. Fei, and P. C. K. Luk, "A new technique of cogging torque suppression in direct-drive permanent-magnet brushless machines," *IEEE Transactions on Industry Applications*, Vol. 46, No. 4, pp. 1332-1340, July/August 2010.
- [5] D. Wang, X. Wang, and S. Y. Jung, "Cogging torque minimization and torque ripple suppression in surface-mounted permanent magnet synchronous machines using different magnet widths," *IEEE Transactions on Magnetics*, Vol. 49, No. 5, pp. 2295-2298, May 2013.
- [6] M. S. Islam, R. Islam, T. Sebastian, A. Chandy, and S. A. Ozsoylu, "Cogging torque minimization in PM motors using robust design approach," *IEEE Transactions on Industry Applications*, Vol. 47, No. 4, pp. 1661-1669, July/August 2011.
- [7] S. Oh, S. Min, and J. P. Hong, "Air gap flux density waveform design of surface-mounted permanent magnet motor considering magnet shape and magnetization direction," *IEEE Transactions on Magnetics*, Vol. 49, No. 5, pp. 2393-2396, May 2013.
- [8] F. Zhao, T. A. Lipo, and B. I. Kwon, "A novel two-phase permanent magnet synchronous motor modeling for torque ripple minimization," *IEEE Transactions on Magnetics*, Vol. 49, No. 5, pp. 2355-2358, May 2013.
- [9] A. Mahmoudi, S. Kahourzade, N. A. Rahim, and W. P. Hew, "Design, analysis, and prototyping of an axial-flux permanent magnet motor based on genetic algorithm and finite-element analysis," *IEEE Transactions on Magnetics*, Vol. 49, No. 2, pp. 1479-1492, April 2013.
- [10] D. K. Woo, I. W. Kim, D. K. Lim, J. S. Ro, and H. K. Jung, "Cogging torque optimization of axial flux permanent magnet motor," *IEEE Transactions on Magnetics*, Vol. 49, No. 5, pp. 2189-2192, May 2013.
- [11] D. K. Lim, D. K. Woo, I. W. Kim, J. S. Ro, and H. K. Jung, "Cogging torque minimization of a dual-type axial-flux permanent magnet motor using a novel optimization algorithm," *IEEE Transactions on Magnetics*, Vol. 49, No. 9, pp. 5106-5111, September 2013.
- [12] C. J. Jian, W. L. Ma, Y. K. Fan, and H. Yang, "New methods for arc permanent magnet linear synchronous motor to decrease torque ripple," *IEEE Transactions on Magnetics*, Vol. 48, No. 10, pp. 2659-2663, October 2012.
- [13] S. Q. A. Shan, T. A. Lipo, and B. I. Kwon, "Modeling of novel permanent magnet pole shape SPM motor for reducing torque pulsation," *IEEE Transactions on Magnetics*, Vol. 48, No. 11, pp. 4626-4629, November 2012.
- [14] D. Flieller, N. K. Nguyen, P. Wira, G. Sturtzer, D. O. Abdeslam, and J. Merckle, "A self-learning solution for torque ripple reduction for nonsinusoidal permanent-magnet motor drives based on artificial neural networks," *IEEE Transactions on Industrial Electronics*, Vol. 61, No. 2, pp. 655-666, February 2014.
- [15] M. Masmoudi, B. E. Badi, and A. Masmoudi, "DTC of B4-inverter-fed BLDC motor drives with reduced torque ripple during sector-to-sector commutations," *IEEE Transactions on Power Electronics*, Vol. 29, No. 9, pp. 4855-4865, September 2014.
- [16] C. Xia, J. Zhao, Y. Yan, and T. Shi, "A novel direct torque control of matrix converter-fed PMSM drives using duty cycle control for torque ripple reduction," *IEEE Transactions on Industrial Electronics*, Vol. 61, No. 6, pp. 2700-2713, June 2014.